

Comparative Study of Multifunctional Biodegradable Bio-Coagulant Beads for Smart Water Treatment Applications

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INTRODUCTION:

Turbidity: Turbidity is an important parameter used to determine water contamination levels. High turbidity affects water clarity, blocks light penetration, and may indicate the presence of harmful pollutants and microorganisms.

Limitations of Conventional System: Conventional purification systems are often expensive, chemical-intensive, and less sustainable for low-resource applications.

The Need for Innovation: Therefore, there is a growing need for eco-friendly, biodegradable, and low-cost water treatment technologies.

PROBLEM STATEMENT:

- **Rising Contamination:**

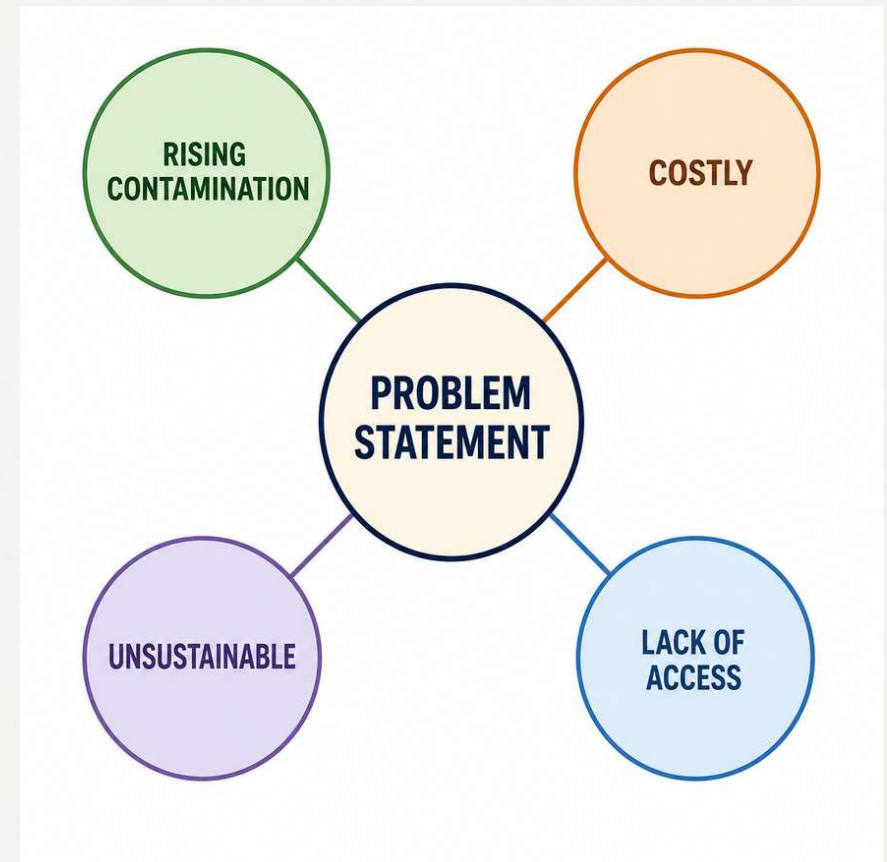
Water contamination due to suspended particles and pollutants is increasing globally.

- **Costly & Unsustainable:**

Conventional purification systems can be expensive and environmentally unsustainable.

- **Lack of Access:**

Conventional purification systems can be expensive and environmentally unsustainable.



AIM OF THE EXPERIMENT:



Reducing Turbidity



Smart Water Quality Monitoring



Improving Water Clarity



Enhancing Coagulation & Adsorption

CORE CONCEPT:

The project utilizes natural biomaterials encapsulated within sodium alginate hydrogel beads to combine three powerful mechanisms:-



Coagulation

Neutralization and settling of suspended particles.



Adsorption

Binding and removal of impurities and contaminants.

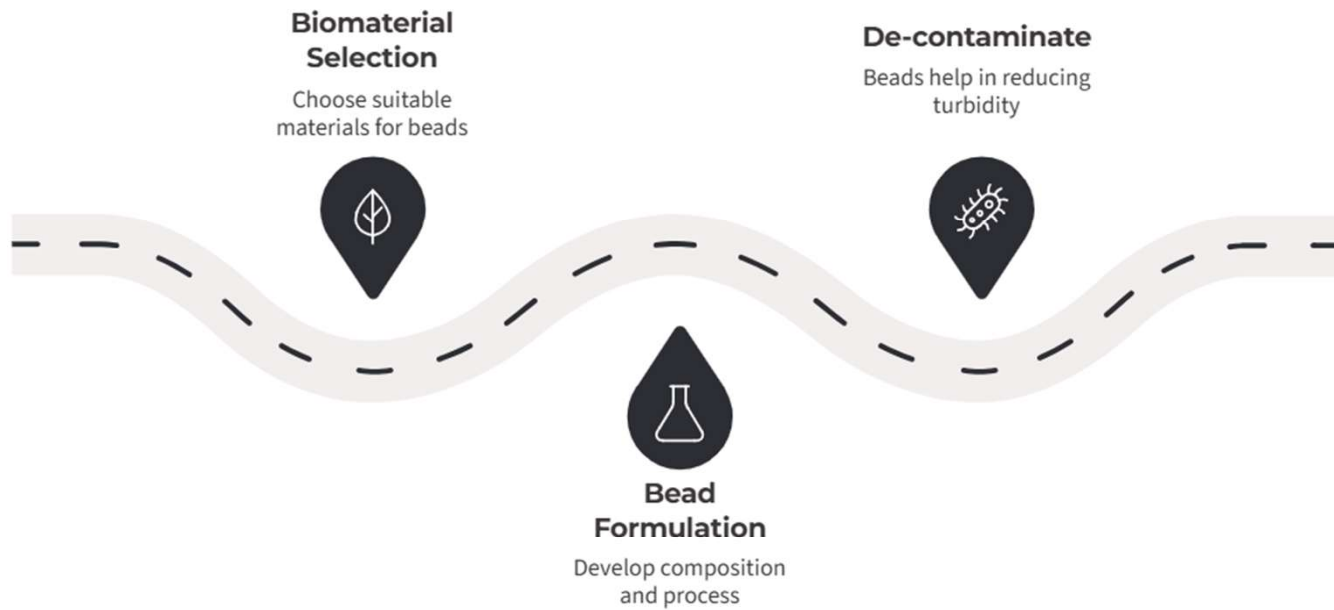


Smart Monitoring

Real-time turbidity analysis using sensors integrated with ESP32.

THE BIOLOGY

INTEGRATION



WHY MORINGA?

Positively charged water-soluble proteins

2S albumin protein

MOCP

- Neutralizes negatively charged suspended particles in dirty water
- Floc formation and coagulation



WHY BANANA PEEL?

Hemicellulose

Pectin

Cellulose

Lignin

- Improves biosorption
- Enhances contaminant binding
- Increases adsorption efficiency



WHY CHARCOAL?

Porous surfaces

- Helps in trapping pollutants, organic impurities, odor molecules and some metal ions.

Adsorption sites



WHY ORANGE PEEL?

Pectin

- Adsorption enhancement
- Pollutant interaction
- Improved purification efficiency

Carboxyl group





HOW WE MADE THE BEADS



Step-by-Step Process (Final Bead – with All Powders)



ROADMAP:

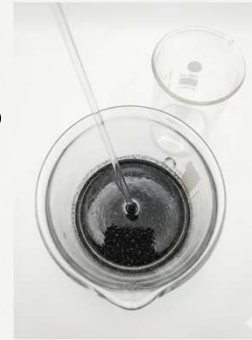
GRINDING:

grinding and weighing of all powders.



GEL MIXTURE:

mix the powders part by part into the gel.



sodium alginate dissolved in water
(double boiling method).

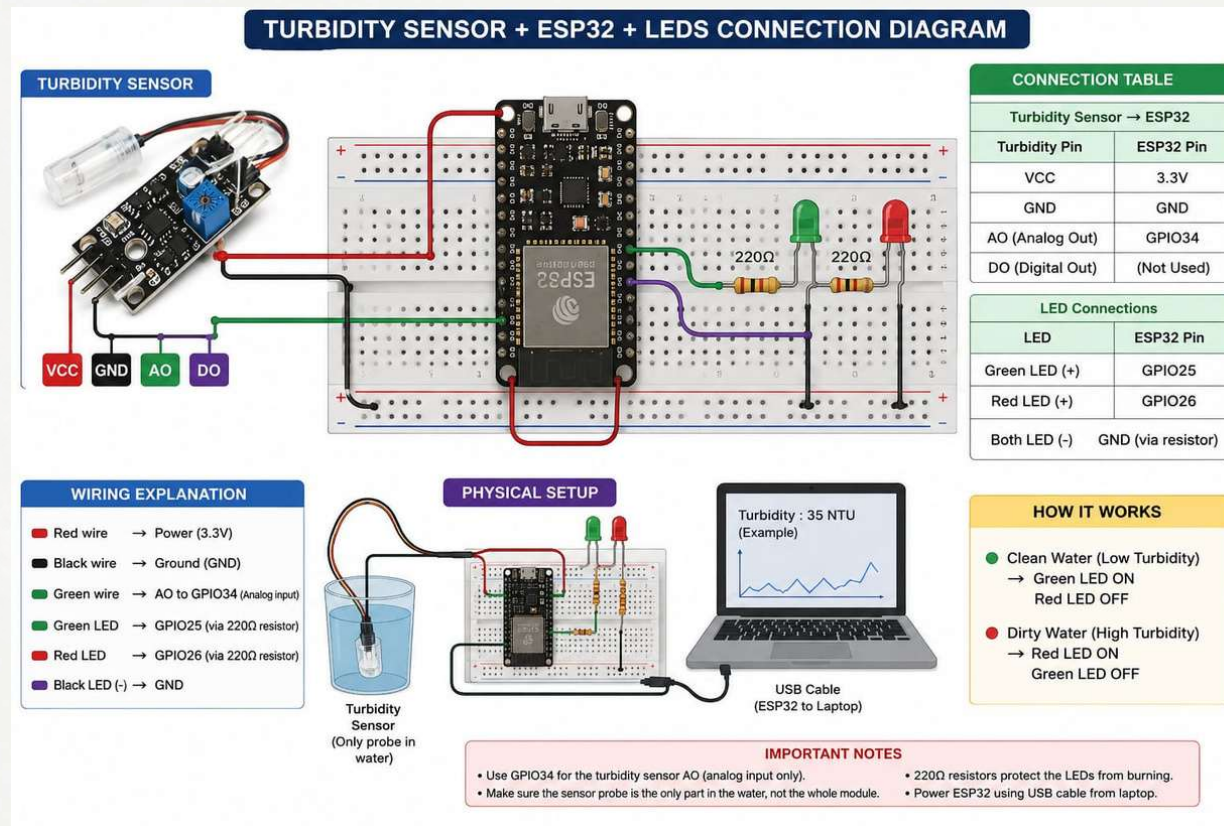
ALGINATE GEL:



drops were dropped into CaCl_2
using a syringe and dried.

BEADS:

THE ET & ISE INTEGRATION



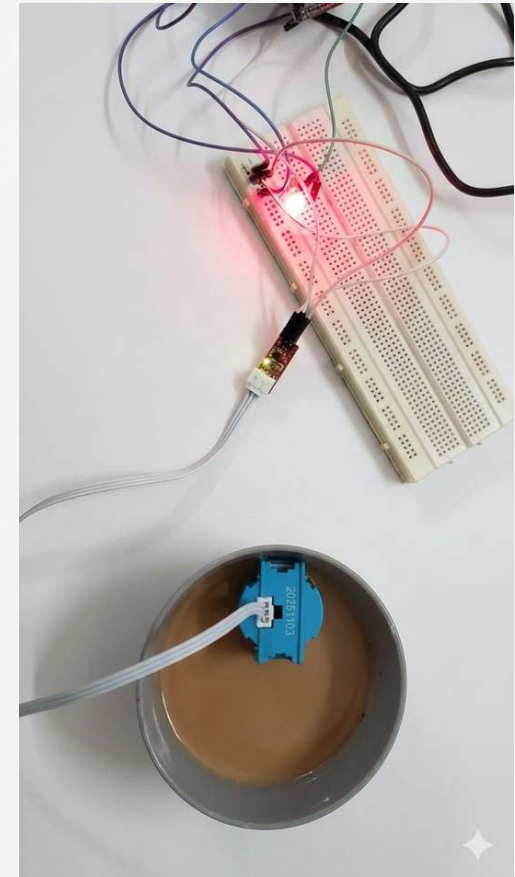
TURBIDITY MONITORING & NTU INTERPRETATION:

- **Sensor Principle:** sensor measures water cloudiness based on light scattering.
- **Data Processing:** ESP32 continuously reads values and converts them into Nephelometric Turbidity Units (NTU) for comparative analysis.
- **NTU VALUE INTERPRETATION:**
 - 0–50 NTU: Clear Water
 - 50–200 NTU: Slightly Turbid Water
 - 200–500 NTU: Moderately Turbid Water
 - Above 500 NTU: Highly Contaminated Water



SIGNIFICANCE OF SMART WATER MONITORING:

- **Continuous Oversight:** Enables consistent, real-time monitoring of water clarity providing instant feedback.
- **Rapid Data-Driven Decisions:** Facilitates instant comparative analysis between raw and treated water, ensuring treatment goals are met efficiently.
- **Validated Performance:** Smart integration confirms particle settling, floc formation, and true turbidity reduction with precise data points rather than subjective visual checks.



ADVANCING SUSTAINABLE SMART WATER

ECOSYSTEM:

- **Low-Cost Implementation:** The use of **affordable** microcontrollers (like ESP32) and **accessible** turbidity sensors demonstrates smart water testing.
- **Sustainable Impact:**
 - Makes **natural, biodegradable** water clean-up methods practical and highly effective.
 - Long-Term Protection:** Supports lasting, easy-to manage environmental monitoring.

ADVANTAGES OF OUR PROJECT:

- The major advantages of our project include:
 - Use of **biodegradable** and eco-friendly biomaterials.
 - Sustainable utilization of agricultural **waste products**.
 - Low-cost and **simple water treatment** approach.
 - Combination of **coagulation** and **adsorption** mechanisms within a single system.
 - **Reduced dependency** on chemical coagulants.
 - Smart **real-time turbidity** monitoring using ESP32 and sensors.



FUTURE SCOPE:

- Heavy **metal adsorption** analysis using advanced laboratory techniques.
- **Industrial wastewater** treatment applications.
- Real-time IoT **cloud monitoring** systems.
- Development of **large-scale purification** units.
- Regeneration and **reuse** studies of hydrogel **beads**.
- **AI** and data analytics integration for **smart water quality prediction**.
- Contaminated water can be made suitable for **drinking** as an upcoming project.



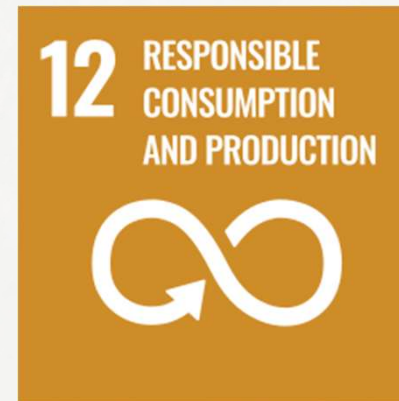
SDG RELEVANCE:



Helps improve access to **cleaner water**, supporting better **public health** and reducing water-related health risks.



The project promotes affordable and sustainable **water purification** methods for improving water quality.



The use of **agricultural waste** materials such as banana peel and orange peel supports **sustainable resource** utilization and waste reduction.



Development of **biodegradable** and **eco-friendly** purification systems helps reduce environmental pollution.

CONCLUSION:

This project demonstrates a sustainable and innovative approach toward water quality improvement using biodegradable materials and smart monitoring technology. The study highlights the potential of eco-friendly solutions in addressing environmental challenges and encourages further development in the field of smart water treatment systems.

THANK YOU!